

Week 6 Cross-Track Integration Evidence

Project: HealthConnect Clinic

Track: Data Analytics

Collaborating Track: Data Science

Data Science Contributor: Shylet

1. Integration Objective

The objective of the Data Analytics–Data Science integration was to provide validated analytical findings that could support the interpretation and improvement of the HealthConnect no-show prediction model.

The Data Analytics findings were reviewed against the Data Science model and error analysis to determine whether the patterns identified in the data were consistent with the model's behaviour and whether any additional relationships should be considered for future modelling.

2. Analytical Findings Provided by Data Analytics

The Data Analytics track provided the following validated findings:

Booking Lead Time

No-show rates increased substantially with longer booking lead times:

- 0–9 days: 29.01%
- 10–19 days: 37.11%
- 20–29 days: 44.44%
- 30–39 days: 50.53%
- 40–49 days: 61.62%
- 50–59 days: 67.92%

This finding was identified as one of the strongest behavioural patterns in the Week 6 analysis.

Previous No-Show History

No-show rates also increased as previous no-show history increased:

- 0 previous no-shows: 43.51%
- 1 previous no-show: 53.49%
- 2 previous no-shows: 59.36%
- 3 previous no-shows: 67.95%

The 4–5 previous-no-show groups were treated cautiously because of their small sample sizes.

Appointment Type

Overall no-show rates were:

- Diagnostic Test: 49.75%
- Follow-up: 51.23%
- General Consultation: 46.64%
- Specialist Consultation: 47.44%

This clarified that Specialist Consultation did not have the highest overall no-show rate.

Reminder Status

- Without reminder: 51.39%
- With reminder: 47.36%
- Difference: 4.03 percentage points

The finding was interpreted as an association rather than proof of causation.

3. Data Science Feedback Received

Data Science reviewed the analytical findings against its model and error analysis.

The Data Science track reported that:

- the baseline Logistic Regression model achieved 62.4% accuracy and 0.677 ROC-AUC;
- the improved tuned Gradient Boosting model achieved 64.8% accuracy and 0.687 ROC-AUC;
- booking lead time was the strongest model feature, accounting for approximately 57% of predictive weight;
- historical no-show rate was the second-strongest feature at approximately 12%.

- These results independently aligned with the Data Analytics findings on booking lead time and previous no-show behaviour.

4. Specialist Consultation Clarification

Data Science initially identified Specialist Consultation as having the highest model error rate, at 42.2%.

Data Analytics clarified that model error rate is different from actual no-show rate.

The Data Analytics results showed:

- Specialist Consultation: 47.44% no-show rate
- Follow-up: 51.23% no-show rate

Therefore, Specialist Consultation did not have the highest actual no-show rate.

This clarification helped Data Science reframe the Specialist Consultation model-error finding as a modelling/performance issue rather than evidence that Specialist Consultation was inherently the highest-risk appointment type.

5. Appointment Type × Booking Lead Time Integration

The Data Analytics segment analysis showed that no-show rates generally increased as booking lead time increased across appointment types.

For example, at 0–9 days:

- Diagnostic Test: 32.47%
- Follow-up: 30.53%
- General Consultation: 26.85%
- Specialist Consultation: 29.22%

At 50–59 days:

- Diagnostic Test: 77.27%
- Follow-up: 72.86%

- General Consultation: 61.90%
- Specialist Consultation: 68.18%

This suggested that appointment type and booking lead time may interact rather than acting as completely independent factors.

Data Science indicated that this interaction could be considered for an explicit interaction feature in Week 7.

6. What Data Analytics Received

The Data Analytics track received:

- model feature-importance information;
- Data Science error-analysis feedback;
- confirmation of the strongest model features;
- clarification of the Specialist Consultation model-error finding;
- and a potential Week 7 modelling direction involving appointment type and booking lead time.

7. What Data Analytics Provided

The Data Analytics track provided:

- validated booking lead-time findings;
- previous no-show findings;
- appointment-type analysis;
- reminder analysis;
- combined relationship analysis;
- validated KPIs;
- segment-level evidence;
- analytical interpretation;
- and cautions concerning small samples and association versus causation.

8. Integration Outcome

The collaboration produced several useful outcomes.

First, the Data Science model independently confirmed the importance of booking lead time and historical no-show behaviour, strengthening confidence in the corresponding Data Analytics findings.

Second, the collaboration corrected the interpretation of the Specialist Consultation result by separating model error rate from actual no-show rate.

Third, the interaction between appointment type and booking lead time was identified as an area requiring further investigation and potential modelling during Week 7.

Therefore, the integration resulted in an actual analytical change and a clear next-step modelling consideration rather than being only a communication exercise.

9. Evidence Available

The following evidence supports the integration:

- Week 6 Data Analytics validation tables.
- Week 6 segment analysis.
- Week 6 relationship analysis.
- KPI validation results.
- Data Analytics response shared with Data Science.
- Data Science's written review and feedback.
- Feature-importance information provided by Data Science.
- Discussion identifying appointment type × booking lead time as a potential Week 7 interaction.
- Cross-track coordination record.

10. Overall Integration Conclusion

The Data Analytics track successfully completed a cross-track integration with Data Science during Week 6.

The exchange provided independent confirmation of key analytical findings, improved interpretation of model performance and identified a specific interaction for further investigation during Week 7.

This integration strengthens the HealthConnect project by connecting validated business analytics with predictive modelling decisions and provides a clear foundation for the next testing and refinement stage.